① (Most Relevant)

6. Identify and describe the mathematical aspects of a real-world problem

✔ **Relevance: ★★★★★**

Analysis:

The task requires students to extract mathematical structure from a real-world context:

- Payment scheme → mathematical relationship
- Number of newspapers sold → independent variable
- Earnings → dependent variable

👉 Students must recognize:

- Piecewise pricing mechanism (first 240 vs. additional units)
- Changing unit rates

📌 Core competence:
Mathematization of real-world situations

---

②

2. Identify the key variables in a model

✔ **Relevance: ★★★★★**

Analysis:

Key variables include:

- x: number of newspapers sold
- $f(x)$: earnings function

Additionally:

- Breakpoint: 240
- Two different rates: 0.20 and 0.40

👉 Incorrect identification of variables leads to failure in modeling

③

9. Translate a problem into a standard mathematical representation

✔ **Relevance: ★★★★★**

Analysis:

The task requires translating verbal information into a formal model:

👉 Piecewise function:

$$f(x) = \begin{cases} 0.20x & x \leq 240 \\ 0.20 \cdot 240 + 0.40(x - 240) & x > 240 \end{cases}$$

📌 This reflects:

- Piecewise function modeling
- Algebraic representation skills

④

7. Simplify or decompose a situation

✔ **Relevance: ★★★★☆**

Analysis:

The situation must be decomposed into:

1. Earnings from the first 240 newspapers
2. Earnings from additional newspapers

👉 This is a typical **structural decomposition process**

Without decomposition, students may incorrectly apply a single rate

---

⑤

4. Read, decode and make sense of statements

✔ **Relevance: ★★★★☆**

Analysis:

The problem includes **text + poster-based information**:

- Two different payment rules
- Linguistically complex descriptions

👉 Students must:

- Correctly interpret "first 240 papers"
- Understand "additional newspapers"

📌 This reflects **information decoding ability**

---

⑥

1. Select a mathematical description

✔ **Relevance: ★★★★☆**

Analysis:

Students must choose:

- A functional representation (rather than simple arithmetic)

👉 This reflects the ability to **select an appropriate model form**

⑦

10. Use mathematical tools to describe relationships

✔ **Relevance:** ★★★☆☆

Analysis:

This includes:

- Constructing expressions
- Substituting values (e.g., 350)

However, the tools required are relatively basic

---

⑧

12. Identify constraints, assumptions

✔ **Relevance:** ★★★☆☆

Analysis:

Implicit assumptions include:

- Sales ≥ 0
- Piecewise rule strictly applies
- No additional costs considered

👉 Reflects **implicit modeling awareness**

---

⑨

5. Recognise mathematical structure

✔ **Relevance:** ★★★☆☆

Analysis:

Students recognize:

- Piecewise function structure
- Linear relationships within segments

However, the structure is relatively simple

⑩

8. Recognise similar known problems

✔ **Relevance:** ★★☆☆☆

Analysis:

Similar to:

- Tiered pricing problems (e.g., electricity bills)

But the task does not explicitly require transfer from prior knowledge

---

⑪

3. Select representation appropriate to context

✔ **Relevance:** ★★☆☆☆

Analysis:

Possible representations:

- Algebraic expressions (primary)
- Graphs (optional, not required)

①（最核心）

6. Identify and describe the mathematical aspects of a real-world problem

✔ **对应性：★★★★★**

分析：

该题首先要求学生从现实情境中抽取数学结构：

- 报酬机制 → 数学函数关系
- 销量（350）→ 自变量
- 收入 → 因变量

👉 学生必须识别：

- 分段收费机制（前240份 vs 超出部分）
- 单位收益变化

📌 本质能力：
现实问题数学化（Mathematization）

---

②

2. Identify the key variables in a model

✔ **对应性：★★★★★**

分析：

关键变量包括：

- x：销售数量
- 收入函数 $f(x)$

并且需要进一步区分：

- 分段点：240
- 两段不同单价：0.20 / 0.40

👉 如果变量识别错误，模型直接失效

③

9. Translate a problem into a standard mathematical representation

✔ **对应性：★★★★★**

分析：

该题核心是将文字转化为：

👉 分段函数模型：

$$f(x) = \begin{cases} 0.20x & x \leq 240 \\ 0.20 \cdot 240 + 0.40(x - 240) & x > 240 \end{cases}$$

📌 这是典型：

- 分段函数建模
- 代数表达能力

④

7. Simplify or decompose a situation

✔️ **对应性：** ★★★★☆

分析：

题目必须被拆解为：

1. 前240份收入
2. 超出部分收入

👉 这是一个典型的“结构分解”过程

否则容易错误地直接用统一单价计算

---

⑤

4. Read, decode and make sense of statements

✔️ **对应性：** ★★★★☆

分析：

题干中信息是**文本+海报形式**：

- 两种不同支付规则
- 不同表达方式（语言复杂）

👉 学生需要：

- 正确理解“first 240 papers”
- 正确理解“additional newspapers”

📌 属于信息解码能力

---

⑥

1. Select a mathematical description

✔️ **对应性：** ★★★★☆

分析：

学生需要选择：

- 用函数表示收入关系
- 而不是简单加减或比例

👉 这是“模型形式选择能力”

⑦

10. Use mathematical tools to describe relationships

✔ **对应性：** ★★★☆☆

分析：

体现为：

- 建立表达式
- 进行计算（代入350）

但工具使用较基础（无复杂符号或图像）

---

⑧

12. Identify constraints, assumptions

✔ **对应性：** ★★★☆☆

分析：

隐含约束：

- 销量 ≥ 0
- 分段规则严格适用
- 不考虑其他费用

👉 属于隐性建模意识

---

⑨

5. Recognise mathematical structure

✔ **对应性：** ★★★☆☆

分析：

识别：

- 分段函数结构
- 线性关系（每段）

但结构较简单，因此权重略低

⑩

8. Recognise similar known problems

✔ **对应性：** ★★☆☆☆

分析：

类似问题：

- 阶梯计费（电费、水费）

但题目未显式要求迁移已有模型

---

⑪

3. Select representation appropriate to context

✔ **对应性：** ★★☆☆☆

分析：

可以用：

- 表达式（主流）
- 图像（非必须）

但题目不强制多表示形式

---

⑫（最低相关）

11. Use diagrams/symbols

✔ **对应性：** ★☆☆☆☆

分析：

虽然有海报，但不是数学图形建模任务